Project 1 Completion Report
Daily Research Knowledge Graph

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1 Project question and current status

Project 1 asks whether a daily agent can collect machine-learning research and organize it as an evidence-backed knowledge graph that researchers can search and explore. The implemented `paper-fetcher-agent` ingests a stored arXiv corpus and enabled public RSS or Atom feeds, extracts grounded features, atomically rebuilds the graph, computes graph and trend analyses, and persists SQLite plus bounded JSON outputs. GitHub Actions schedules the live collector each day at 13:17 UTC and supports manual runs [workflow].

The complete path now runs from collection through a browser interface. The active graph is deterministic under the rules extractor and every semantic edge records either extractor evidence or source metadata. Remaining work is evaluation rather than basic implementation: summary quality and graph placement have not been judged by human researchers, public-feed coverage is bounded, and the daily arXiv job currently queries only `cs.LG`.

1.1 Verified August 6 snapshot

Measure	Count	Measure	Count
Paper nodes	7,794	Report nodes	29
Blog-post nodes	906	Social-post nodes	25
Document nodes	8,754	Concept/category nodes	197
Total nodes	8,951	Typed edges	59,256

The active snapshot combines 100 Google DeepMind items, 835 Hugging Face items, and 25 public Reddit `r/MachineLearning` posts with the stored arXiv corpus. Every one of the 8,754 document nodes has at least one edge. The database is `data/arxiv_kg.sqlite3`; bounded analysis artifacts are in `output/knowledge_graph/` [run summary].

1.2 Requirement coverage

Project requirement	Observed implementation
Collect daily	Scheduled workflow plus manual dispatch.
Use several sources	arXiv, research-blog/report feeds, and one public social feed.
Build knowledge graph	Four document types and six concept/category types.
Explain relationships	Typed edges retain evidence or explicit metadata basis.
Find structure	Search, placement, hubs, clusters, hot topics, and emerging topics.
Recover safely	Transactional source checkpoints and atomic graph replacement.

The snapshot contains no isolated document nodes. This demonstrates structural coverage inside the stored corpus, not scientific correctness or importance. Those stronger claims require human evaluation.

2 Collection, normalization, and feature extraction

2.1 Inputs and source boundaries

The agent starts from version-aware arXiv records already stored in the repository and checks configured live sources. Feed adapters normalize each item into a common schema containing source identity, stable item identifier, URL, title, summary text, publication and update timestamps, retrieval time, content hash, and raw provenance. Report-like Hugging Face titles become `report` nodes; other Hugging Face and DeepMind entries become `blog_post` nodes. Reddit entries become `social_post` nodes.

Source	Stored role	Items	Boundary
Stored arXiv corpus	papers	7,794	Multi-disciplinary historical corpus
Google DeepMind	research blog	100	Public RSS or Atom output
Hugging Face	blog/report	835	Public feed; 29 report-like titles
Reddit r/MachineLearning	social media	25	One public subreddit feed

Public feeds do not represent all blogs, reports, or social media. The workflow does not use authenticated or private sources. The tracked run summary has `arxiv: null`, meaning that particular snapshot proves feed collection and graph construction, not a fresh arXiv fetch during that exact run.

2.2 Daily pipeline

Step	Operation	Output or safety rule
1	Bootstrap stored papers	Stream JSONL; keep canonical versioned records.
2	Collect enabled sources	Enforce response, redirect, item-count, and time bounds.
3	Normalize and upsert	Store content hash and provenance; update idempotently.
4	Extract missing features	Version feature rows by extractor and prompt versions.
5	Build graph	Validate node identities and edges before atomic replacement.
6	Analyze and export	Write hubs, clusters, trends, placement data, and summary JSON.

2.3 Grounded feature extraction

Rules extractor 1.4 produced 7,794 current paper feature rows. It recognizes a documented finite vocabulary for topics, methods, research goals, datasets, metrics, contributions, and limitations. Exact text evidence accompanies semantic values. Related-work and negation checks reduce false attributions; unsupported free keywords never create semantic graph edges.

Summaries copy the first sentence of an abstract or feed summary and retain source plus extractor provenance. This avoids a generated model inventing a new factual claim, but extractive does not automatically mean useful or complete. No human-rated summary-quality study has been run.

2.4 Pipeline pseudocode

```
1.  stream stored papers into SQLite
2.  for each enabled source:  fetch, validate, normalize, upsert, checkpoint
3.  extract only papers missing current extractor version
4.  build candidate nodes and evidence-backed typed edges
5.  validate counts and references, then replace active graph atomically
6.  compute bounded analyses and write JSON artifacts
```

3 Knowledge graph, analysis, and interface

3.1 Graph schema and active counts

Document nodes represent papers, reports, blog posts, and social posts. Concept nodes represent arXiv categories, topics, methods, research goals, datasets, and metrics. ArXiv categories and configured feed topics use source metadata. Other semantic relationships require exact extractor evidence.

Node type	Count	Node type	Count
Paper	7,794	Category	148
Report	29	Dataset	11
Blog post	906	Method	11
Social post	25	Metric	10
Topic	7	Research goal	10

Relation	Count	Meaning
`RELATED_TO`	37,656	Supported-concept overlap between documents
`IN_CATEGORY`	13,180	Recorded arXiv category membership
`ABOUT_TOPIC`	3,056	Evidence or configured source topic
`PURSUES_GOAL`	1,972	Extracted research objective
`USES_METHOD`	1,636	Extracted method evidence
`REPORTS_METRIC`	1,624	Extracted reported metric
`EVALUATES_ON`	132	Extracted dataset evidence

3.2 Relationships, hubs, and clusters

`RELATED_TO` uses IDF-weighted overlap among supported concepts. Edges need score at least 0.20, and each document has at most ten related-document neighbors. This degree cap keeps a common topic from turning the graph into one dense component. Hub analysis reports weighted PageRank, degree, and weighted degree. These measure graph centrality, not research quality.

Clusters are connected components over `RELATED_TO` edges scoring at least 0.35. Labels come from concepts supported inside each component. This is a deterministic graph grouping, not a learned scientific taxonomy [system report].

3.3 Hot and emerging topics

Trend analysis compares equal recent and baseline windows using mean per-source document shares, which prevents a high-volume source from dominating raw counts. Emerging score requires at least three recent documents and positive smoothed log growth. Seven-day windows ending August 6 have status `ok`; no topic met both conditions in the active mixed-source snapshot [system report].

3.4 Search and exploration interface

Local browser UI combines document search, graph-placement explanation, hot topics, emerging topics, hubs, and clusters. Five bounded JSON APIs expose the same data. Placement output names the selected document, typed relationships, and neighboring nodes, so a user can inspect why it appears in one graph area. Integration tests cover empty graph state, placement, active build metadata, and API bound validation. Active build ID is `95ac2bfcd067acf6cf2b7ce03fd686920ebfb14a95b3364d0c38095bcbae00c2`.

4 Reliability, limitations, and next work

4.1 Persistence and recovery

Each public feed stores normalized items and its success checkpoint in one SQLite transaction. A failure therefore cannot commit a checkpoint without its corresponding items. Graph replacement is also atomic: validation occurs before the active build changes, and a failed rebuild leaves the prior graph and analyses readable. Tracked database integrity is `ok`, and the foreign-key check returns no violations.

The first August 6 arXiv smoke run found 3,021 submissions beyond the configured 200-paper safety cap and stopped without advancing its checkpoint. Because this project intentionally keeps a bounded corpus, the two `cs.LG` checkpoints were rebased to August 6. Papers not already stored from July 13 through the rebase time were deliberately skipped. A later 24-hour overlap run completed with zero new submissions and a valid checkpoint transition [system report].

4.2 Relationship to category classification

The repository also contains a separate neural classifier trained on 7,708 eligible papers across 112 primary arXiv categories. Its held-out test set has 1,157 papers; observed accuracy is 61.37%, macro-F1 is 34.65%, weighted-F1 is 59.15%, and top-3 accuracy is 84.62%. Those predictions are not graph ground truth and do not create graph edges. Graph placement instead uses recorded arXiv categories, configured source metadata, and rule-extractor evidence. The classifier's architecture, errors, and limitations remain in the separate four-page classification report.

4.3 Interpretation limits

Public feeds do not cover every blog or social platform; Reddit input comes from one public subreddit. The rules extractor has a finite vocabulary, so a missing edge can mean missing evidence or an unrecognized concept. No human-rated study has checked summary usefulness or graph placement. An absent edge can mean absent evidence or an unrecognized term. Trend scores describe movement inside this corpus; they do not establish research quality, causal importance, whole-field coverage, or future impact.

4.4 Measurable next steps

1. Have two domain researchers rate a stratified sample of 100 summaries and 100 placement explanations for factual support and usefulness.
2. Expand the rules vocabulary only for concepts with documented false negatives, then rerun the same placement audit.
3. Add another independent public research feed and compare source-normalized trend stability before and after the change.
4. Run the scheduled agent for 30 days and report per-source success, failure, and checkpoint-recovery counts before expanding collection volume.

4.5 Conclusion

Project 1 now provides a reproducible daily research graph with 8,754 documents, 8,951 nodes, 59,256 typed edges, and no isolated document nodes. Its strongest property is traceability: semantic placement is tied to text evidence or source metadata, and operational state is stored transactionally. Current results show a working research-navigation system inside a bounded corpus. They do not yet prove human-level summary accuracy, scientifically correct placement, or complete coverage of machine-learning research.

Repository evidence. [system report] [run summary] [workflow] [classification report source]